



TECHNICAL PROFESSIONAL

ICT - COMPUTER PROGRAMMING, ORACLE DATABASE

Grade 12

**Three-Term Budget Of Work (BOW)
for Learning Competencies**

TECHNICAL PROFESSIONAL
ICT - ICT SUPPORT AND COMPUTER PROGRAMMING TECHNOLOGIES
Grade 12

Three-Term Budget Of Work
For Learning Competencies

Please note that each TechPro elective is completed within one term if delivered at Grade 12.

Term Plan	
Elective	Computer Programming, Oracle Database

Week	Learning Competency
*	• Discuss fundamental concepts and components of Oracle Database system. • Discuss installation procedure of IDE for Oracle Database. • Construct basic SQL syntax. • Apply LIMIT rows syntax retrieved by a query. • Apply SORT rows syntax retrieved by a query. • Apply single row functions. • Apply multiple row functions. • Apply JOIN syntax in retrieving and displaying from multiple tables. • Apply subqueries for complex data retrieval and manipulation. • Apply substitution variables in SQL.
*	• Apply DML commands in manipulating data for table attributes. • Create tables for managing data.

Week	Learning Competency
	• Apply constraint commands for data validation. • Apply VIEW syntax in accessing specific data. • Apply database object functions for storing or referencing data. • Apply database control in managing user access. • Apply SET operators in data manipulation. • Execute GROUP BY clause commands for data manipulation.
*	• Perform connection of PL to SQL for program application. • Apply SQL syntax to programming .net or Java for storing or retrieval of data. • Evaluate bugs using exception handling in programming .net or Java for debugging process. • Apply store procedures on PL/SQL. • Create the PL/SQL code framework. • Apply the advanced interface methods on PL/SQL framework. • Apply performance and tuning using PL/SQL code.

* Teachers may assign learning competencies per week depending on the needs of the learners.